

DELHI, ON, Canada

WMO: 715730

Lat: 42.8635N

Lon: 80.5453W

Elev: 232

StdP: 98.57

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-16.4	-23.2	0.5	-18.7	-19.8	0.7	-15.9	8.4	-1.6	7.7	-1.4	1.6	260	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	30.7	21.9	29.3	21.3	27.9	20.6	23.9	27.9	23.0	26.9	22.1	25.9	3.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	25.9	21.7	16.9	25.1	20.9	16.0	24.2	73.1	28.1	69.3	26.9	65.9	26.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.5	5.8	-24.3	32.8	4.7	1.7	-27.7	34.0	-30.5	35.0	-33.2	35.9	-36.6	37.1		
			-24.6	25.6	4.6	1.4	-27.9	26.6	-30.6	27.4	-33.2	28.2	-36.5	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-5.2	-4.5	0.4	7.0	13.7	18.8	21.1	20.1	16.5	10.1	4.2	-1.7
	DBStd	10.45	6.50	5.97	6.02	4.87	4.62	3.55	3.06	2.87	3.89	4.57	4.71	5.26
	HDD10.0	1913	471	406	302	114	16	0	0	0	2	57	182	362
	HDD18.3	3866	728	639	555	340	158	36	9	15	80	259	425	620
	CDD10.0	1347	1	0	5	25	130	264	344	313	198	59	7	1
	CDD18.3	258	0	0	0	1	14	51	94	69	26	3	0	0
	CDH23.3	2316	0	0	2	9	167	466	866	557	233	16	0	0
	CDH26.7	578	0	0	0	0	36	122	249	118	52	1	0	0
Wind	WSAvg	2.4	2.8	2.7	2.6	2.6	2.3	2.2	2.1	1.9	1.9	2.3	2.5	2.6
Precipitation	PrecAvg	972	59	56	82	90	79	82	76	91	89	76	97	89
	PrecMax	1197	120	124	193	152	141	167	185	266	291	162	199	146
	PrecMin	629	18	15	12	39	26	15	33	10	20	12	31	37
	PrecStd	130	25	29	41	29	36	33	36	55	57	36	40	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	12.6	20.5	24.7	29.7	31.5	32.9	31.7	30.5	25.4	18.3	13.9
		MCWB	11.7	9.8	15.3	16.7	19.7	21.4	23.2	23.1	21.8	19.0	13.7	12.0
	2%	DB	9.3	9.0	15.2	21.6	27.5	29.8	31.0	29.4	28.1	22.3	15.8	10.2
		MCWB	7.8	7.3	10.7	13.9	18.5	20.9	22.4	22.1	21.3	17.3	12.7	8.6
	5%	DB	5.7	5.8	12.1	18.6	25.2	28.0	29.4	27.8	26.0	20.1	14.0	7.6
		MCWB	4.3	3.9	9.1	12.4	17.7	20.6	21.4	21.0	20.0	16.0	11.3	6.4
	10%	DB	3.0	3.0	9.3	15.8	22.5	26.0	27.8	26.4	24.0	17.9	11.9	5.1
		MCWB	2.0	1.5	6.5	10.7	15.7	18.8	20.6	20.5	18.9	14.6	9.5	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	10.9	15.6	18.7	21.6	23.8	25.7	25.2	24.0	21.0	15.3	11.8
		MCDB	13.2	11.9	20.6	22.3	26.8	29.3	30.0	28.8	28.0	24.0	16.9	13.3
	2%	WB	8.0	7.3	12.0	15.5	20.1	22.5	24.2	23.7	22.6	18.6	13.4	9.0
		MCDB	8.9	9.1	14.3	19.2	25.7	27.5	28.5	27.0	26.0	21.0	15.6	9.8
	5%	WB	4.4	4.0	9.3	13.5	18.7	21.4	23.0	22.7	21.2	16.6	11.5	6.4
		MCDB	5.4	5.6	11.6	17.5	23.4	26.0	26.9	26.1	24.3	19.1	13.3	7.5
	10%	WB	2.1	1.7	6.6	11.3	17.1	20.3	22.0	21.7	19.8	15.0	9.7	4.1
		MCDB	2.9	3.0	9.0	15.1	21.0	24.4	26.1	24.9	23.0	17.3	11.7	5.0
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	9.0	11.0	11.8	11.2	11.6	11.3	11.4	9.5	8.1	6.6
		MCDBR	9.2	11.1	13.6	16.2	14.8	13.3	13.0	12.7	13.9	12.6	11.6	9.1
		MCWBR	8.6	9.4	9.8	10.0	7.5	6.3	6.1	6.5	7.4	7.8	8.8	8.1
	5% WB	MCDBR	8.7	11.0	12.5	13.2	13.0	11.5	10.8	10.3	11.1	10.7	10.4	8.9
		MCWBR	8.5	9.6	9.8	9.6	7.6	6.5	5.7	6.0	6.7	7.7	8.5	8.2
Clear-Sky Solar Irradiance	taub	0.294	0.342	0.334	0.372	0.397	0.430	0.422	0.407	0.379	0.358	0.313	0.291	
	taud	2.396	2.202	2.366	2.320	2.298	2.235	2.279	2.307	2.369	2.410	2.504	2.474	
	Ebn at Noon	845	842	908	894	877	846	848	850	849	821	820	820	
	Edh at Noon	78	112	108	122	128	137	130	122	107	91	71	67	
All-Sky Solar Radiation	RadAvg	1.48	2.13	3.41	4.51	5.55	6.18	6.13	5.37	4.24	2.59	1.59	0.99	
	RadStd	0.13	0.28	0.40	0.48	0.52	0.40	0.42	0.27	0.39	0.24	0.23	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-0.80	-1.15	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page